

conclusion

MB in the brain

Markov Blanket

Markov processes

Dynamical systems

Complex systems

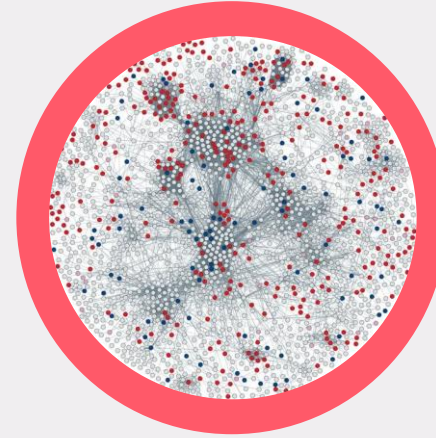
Markov Blanket

FREE ENERGY PRINCIPLE

Shirin Ahmadi

Allostasis Research Group





Complex Systems

complex systems, are systems that consist of a large number of interconnected components or agents that interact with each other in non-linear ways. These systems are characterized by their unpredictability and the emergence of complex behaviors from relatively simple underlying rules.



Different types of complex systems

Biological Systems

- Ecosystems
- The human brain

Physical systems

- Weather systems
- Fluid dynamics

Economic systems

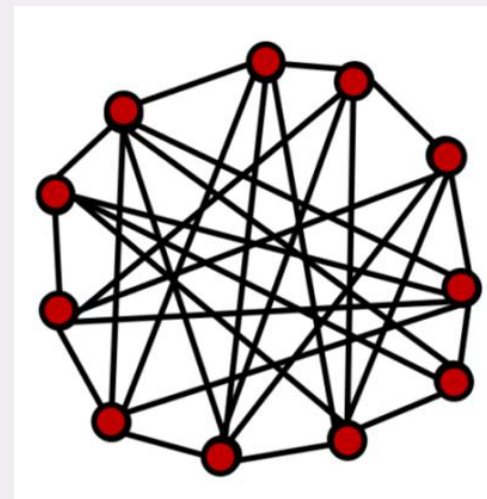
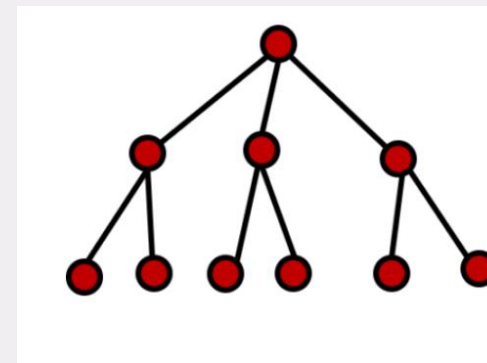
Social systems



Characteristics of complex dynamic systems



- Causal Opacity
- Self Organizing
- Homeostasis
- Coevolution
- Emergence
- Spontaneous order



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Complex systems

In complex systems, we often encounter various processes that describe how different aspects of the system change or evolve over time. These processes are used to model and understand the behavior of complex systems. Here are some commonly used processes:

1

**Bernoulli
process**



2

**Poisson
process**



3

**Markov
process**



4

**Historical
process**



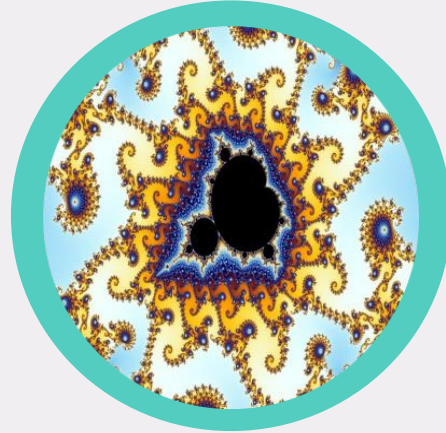
Complex systems

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Dynamical systems

Stochastic
Deterministic

How a system changes or evolves
over time.



Dynamical systems

Complex systems

Deterministic Dynamical Systems

$$i \rightarrow j = \varphi(i)$$

Stochastic Dynamical Systems

- A type of stochastic process commonly studied in probability theory and stochastics is called Markov process
- the future behavior of the process, at any given time, is solely determined by its current state and is independent of how it arrived at that state.
- This makes Markov processes particularly useful for modeling systems or processes that exhibit this memoryless behavior, as it simplifies the modeling and analysis of their behavior.

$$x_t$$

$$P(x_{t+1}|x_0, x_1, x_2, \dots, x_t) = p(x_{t+1}|x_t)$$
$$P(x_{t+1} = j|x_t = i) = p_{ij}$$



Stochastic and Chaos?



Random	Deterministic	
Stochastic	- Low number of parameters, - Exhibit low complexity, - Are not highly sensitive to initial conditions	Predictable
	Chaotic	Unpredictable

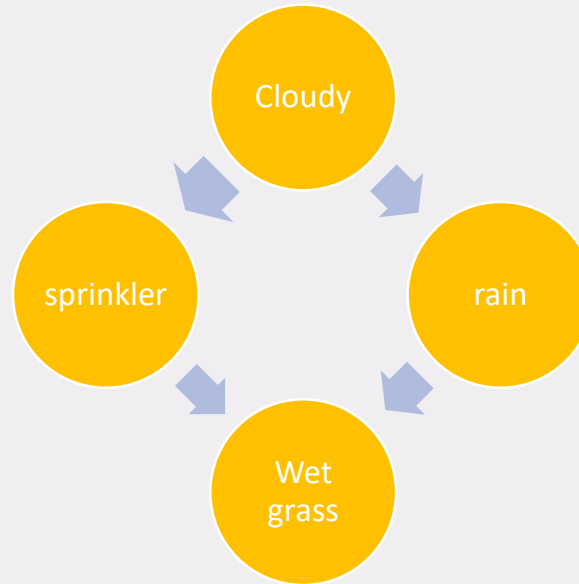


Markov processes

Markov process," which is a way to understand random events by assuming that what will happen in the future only depends on what's happening right now and not on what happened in the past.



Bayesian Network



Markov relations in a Bayesian network can be related in various ways, and these relationships help determine the conditional independence or dependence of variables within the network.

Here are different ways in which Markov relations can be related:



Markov processes

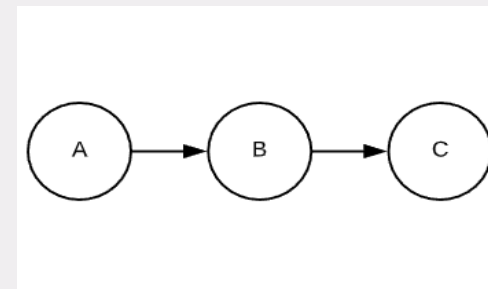
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Complex systems



Causal chain

- If you know everything about B, then knowing the state of A doesn't give you any extra information about C, and vice versa.
- In other words, if you have B, you already have all the information you need to understand what's going on with both A and C.
- The state of A does not contribute additional info to C if B is already known.



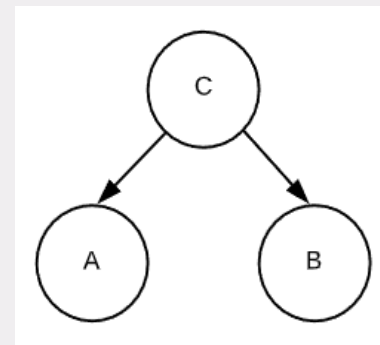
$$P(C|B, A) = P(C|B)$$





Common Cause

- In common cause structures, multiple child nodes share a common parent node.
- In the absence of any prior knowledge, the child nodes are conditionally dependent.
- In common causes, multiple child nodes share the same parent, resulting in conditional dependence among the children.
- For instance, if it is snowing outside, it's reasonable to predict that tomorrow will be windy because both events are influenced by the common cause, which is likely the winter season.
- However, when we have certain knowledge about the common cause (C), such as knowing it is winter, A and B become conditionally independent.
- This means that within the winter season itself, the chances of getting either a windy or snowy day are uncorrelated.



$$P(A|C, B) = P(A|C)$$
$$P(B|C, A) = P(B|C)$$

A and B become
conditionally
independent when
the common cause C
is known

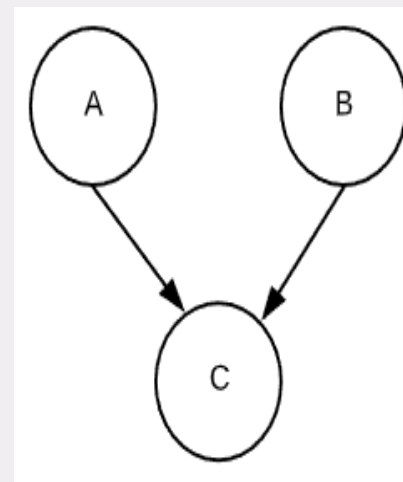




Common Effect

without knowing anything specific about C, A and B are not connected. (For example, if a doctor is trying to diagnose a patient, they might have several ideas about what's causing the patient's symptoms, like inflammation, flu, or indigestion. At this point, the causes (A and B) are not connected because the doctor doesn't have enough information about the patient's symptoms (C).)

once you know the effect (C), it can make the causes (A and B) become dependent on each other. (However, here's the interesting part: when the doctor learns more about the patient's symptoms, like finding out that the patient has a fever, suddenly the causes (A and B) become related. If the patient has the flu (B), it's more likely to explain the fever, so the probability of inflammation (A) goes down. In other words, once you know the effect (C), it can make the causes (A and B) become dependent on each other.)



$$P(A|B) = P(A)$$
$$P(B|A) = P(B)$$

In the absence of knowledge about C, A and B are independent. In the presence of C, they are dependent (not shown)



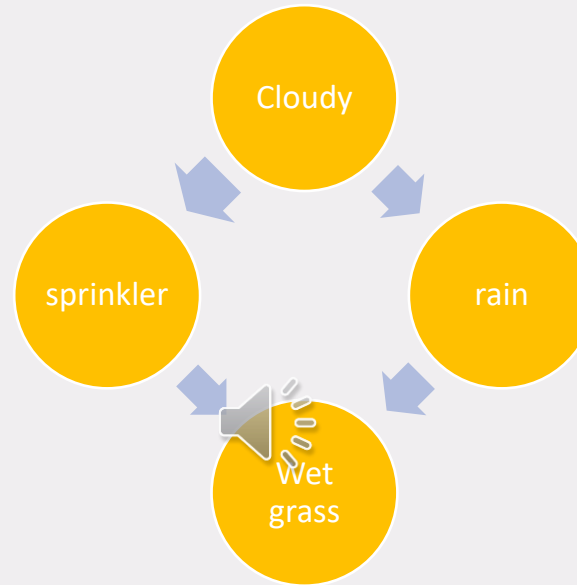


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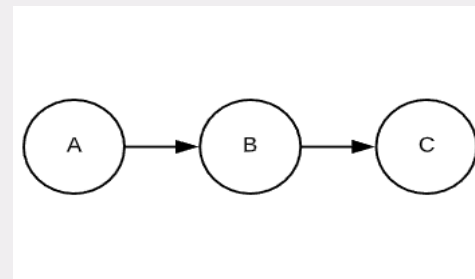
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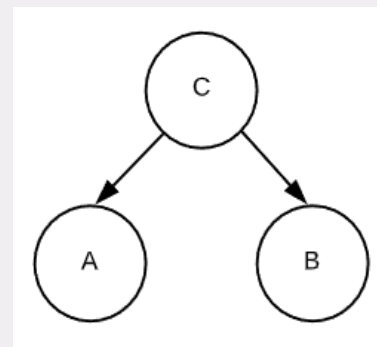
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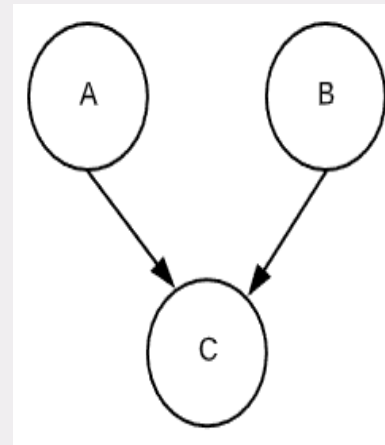


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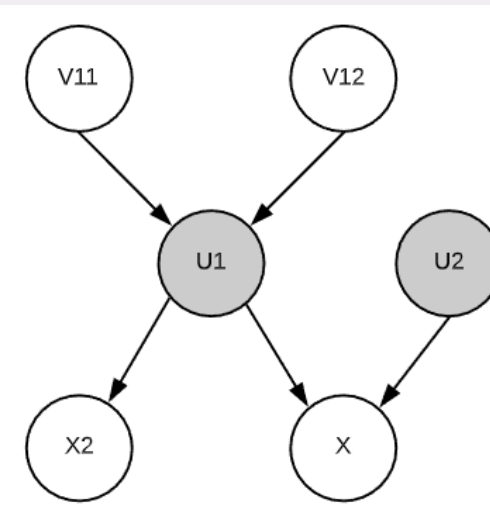
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In the absence of knowledge about C, A and B are independent. In the presence of C, they are dependent (not shown)



1.First, we see the setup $V \rightarrow U \rightarrow X$, which is similar to a causal chain. This means that X depends on its immediate parents, $U1$ and $U2$, but it's independent of anything further upstream due to the Markov property. So, X doesn't depend on anything beyond $U1$ and $U2$.

2.Next, we have $X2 \leftarrow U1 \rightarrow X$, which is similar to a common cause situation. But here's the twist: since we already know $U1$, X and $X2$ become conditionally independent. In other words, knowing $U1$, we don't need to worry about the relationship between X and $X2$ anymore.



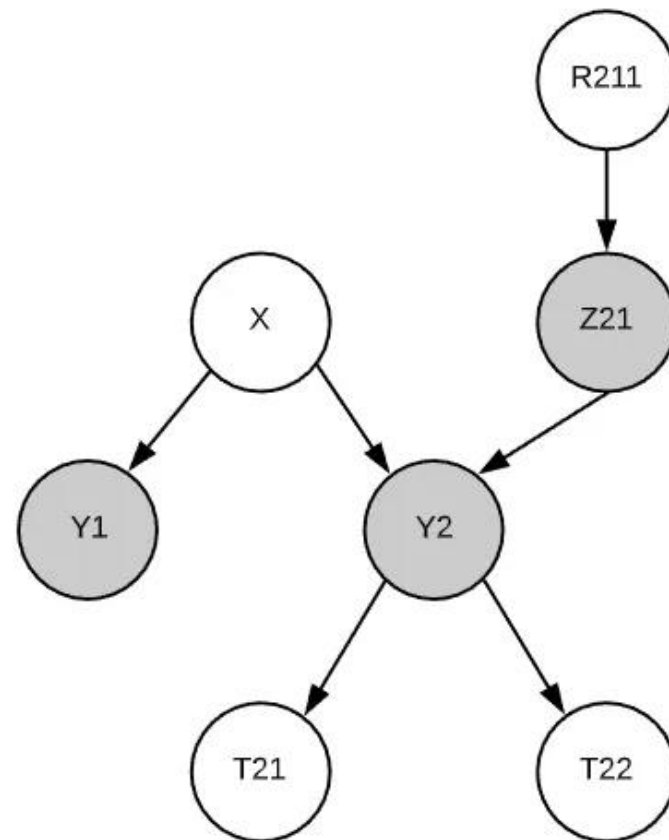
Hence, we know that only the states of $\text{Parents}(X)$ are part of what's needed to evaluate $P(X)$.





$P(X)$ is dependent upon $Y1, Y2$ by Bayes' rule ($P(A|B) = \alpha P(B|A)$). The setup $X \rightarrow Y2 \leftarrow Z21$ is that of (iii) common effect, and since $Y2$ must be known, X and $Z21$ become conditionally dependent. Thus, we know that $\text{Children}(X)$ and $\text{Parents}(\text{Children}(X))$ are both necessary factors in calculating $P(X)$.

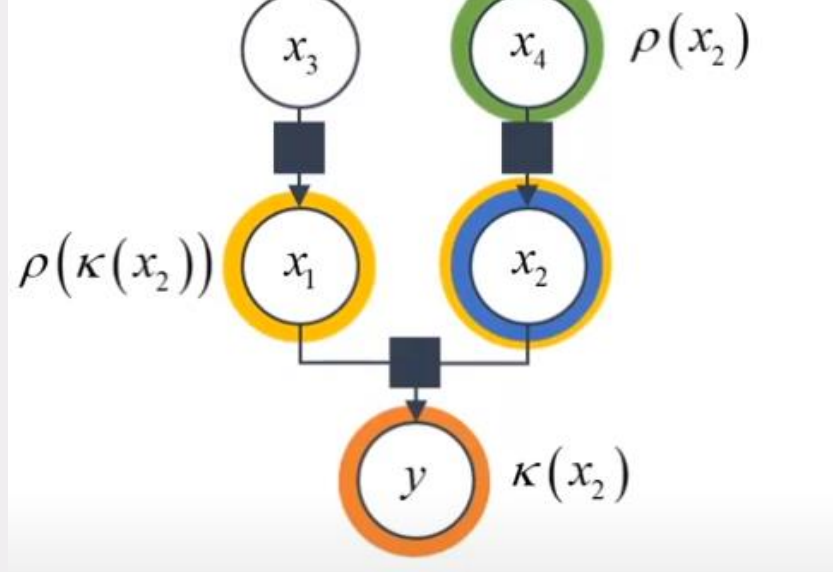
In totality, $P(X)$ is dependent upon $\text{Parents}(X)$, $\text{Children}(X)$ and $\text{Parents}(\text{Children}(X))$, and once all of them are known, independent of every other factor in the Bayes net.



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Transforming a process into a Markov chain



Suppose that the weather in a city is modeled as follows:

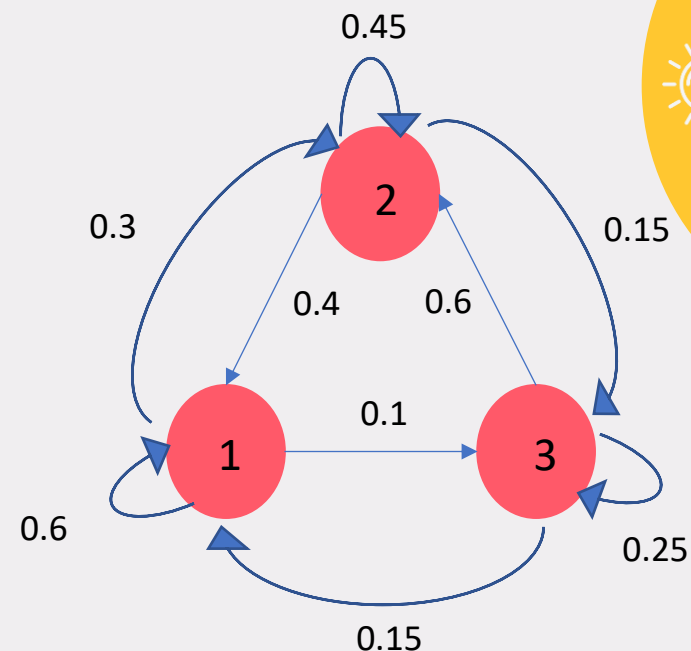
1. If it is sunny today, there is a 60% probability that it will be sunny tomorrow, a 30% chance of being cloudy, and a 10% chance of rain.

2. If today is cloudy, there is a 40% chance that it will be sunny tomorrow, a 45% chance of it being cloudy tomorrow, and a 15% probability of rain tomorrow.

3. If it is rainy today, there is a 15% chance that it will be sunny tomorrow, a 60% chance that it will be cloudy tomorrow, and a 25% chance of rain tomorrow.

Now, given that it is rainy today (Friday), we want to determine the weather forecast for Monday.

1- sunny
2- cloudy
3- rainy



Probability distribution of states

$$\pi(\text{Friday}) = \pi(0) = [0 \ 0 \ 1]$$

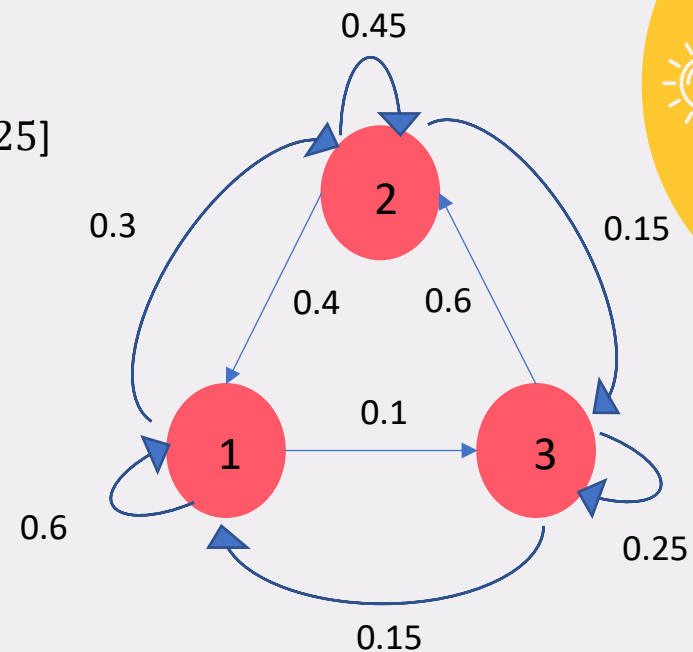
$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.4 & 0.45 & 0.15 \\ 0.15 & 0.6 & 0.25 \end{bmatrix}$$

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$$\pi(1) = \pi(0)P = [0 \ 0 \ 1] \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.4 & 0.45 & 0.15 \\ 0.15 & 0.6 & 0.25 \end{bmatrix} = [0.15 \ 0.6 \ 0.25]$$

$$\pi(3) = \pi(2)P = [0.43 \ 0.42 \ 0.14]$$

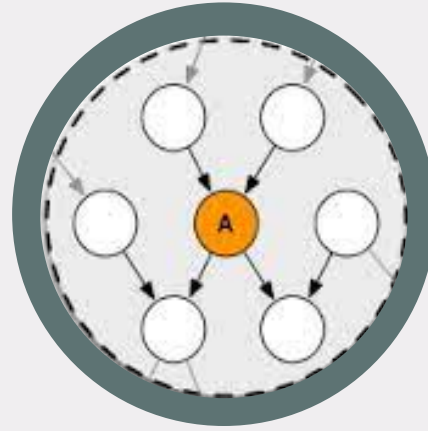
1- sunny
2- cloudy
3- rainy



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Markov Blanket

A MB is a statistical tool that allows explaining how an embodied system, situated in its environment, maintains itself by engaging in dynamics of exchanging with its environment

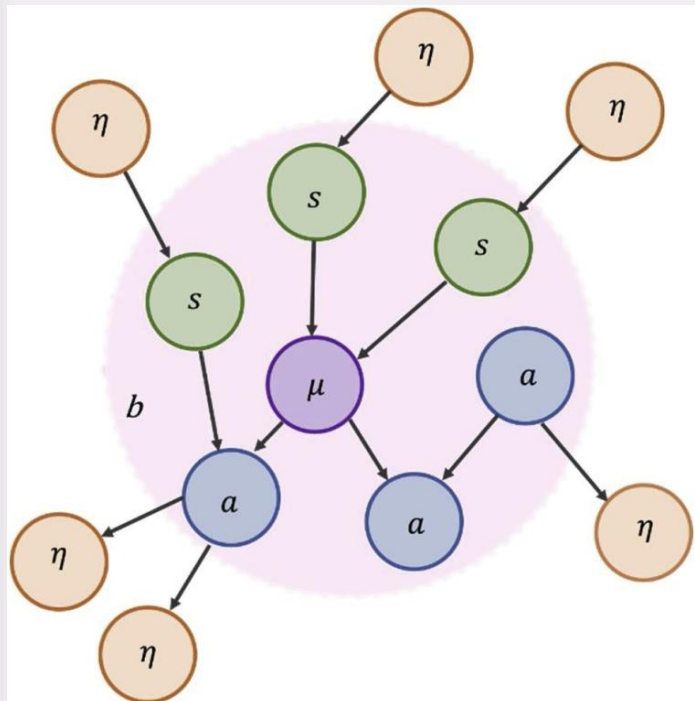


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Markov blanket is not necessarily a physical boundary such as a cell membrane, but rather a statistical one defined by variables that are conditionally independent of each other.

A Markov blanket is like a protective shield that separates a system from its surroundings. This shield ensures that certain variables inside the system (denoted as η) are not influenced by other variables inside the system (denoted as μ). This separation happens because of the presence of the Markov blanket (represented as 'b').



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The conditional dependency structure of Eq.



$$\mu \perp \eta | b \Leftrightarrow p(\mu, \eta | b) = p(\mu | b) p(\eta | b)$$

The dynamics of Eq.



$$\dot{\mu} = f_{\mu}(\mu, s, a)$$

$$\dot{a} = f_a(\mu, s, a)$$

$$\dot{\eta} = f_{\eta}(\eta, s, a)$$

$$\dot{s} = f_s(\eta, s, a)$$

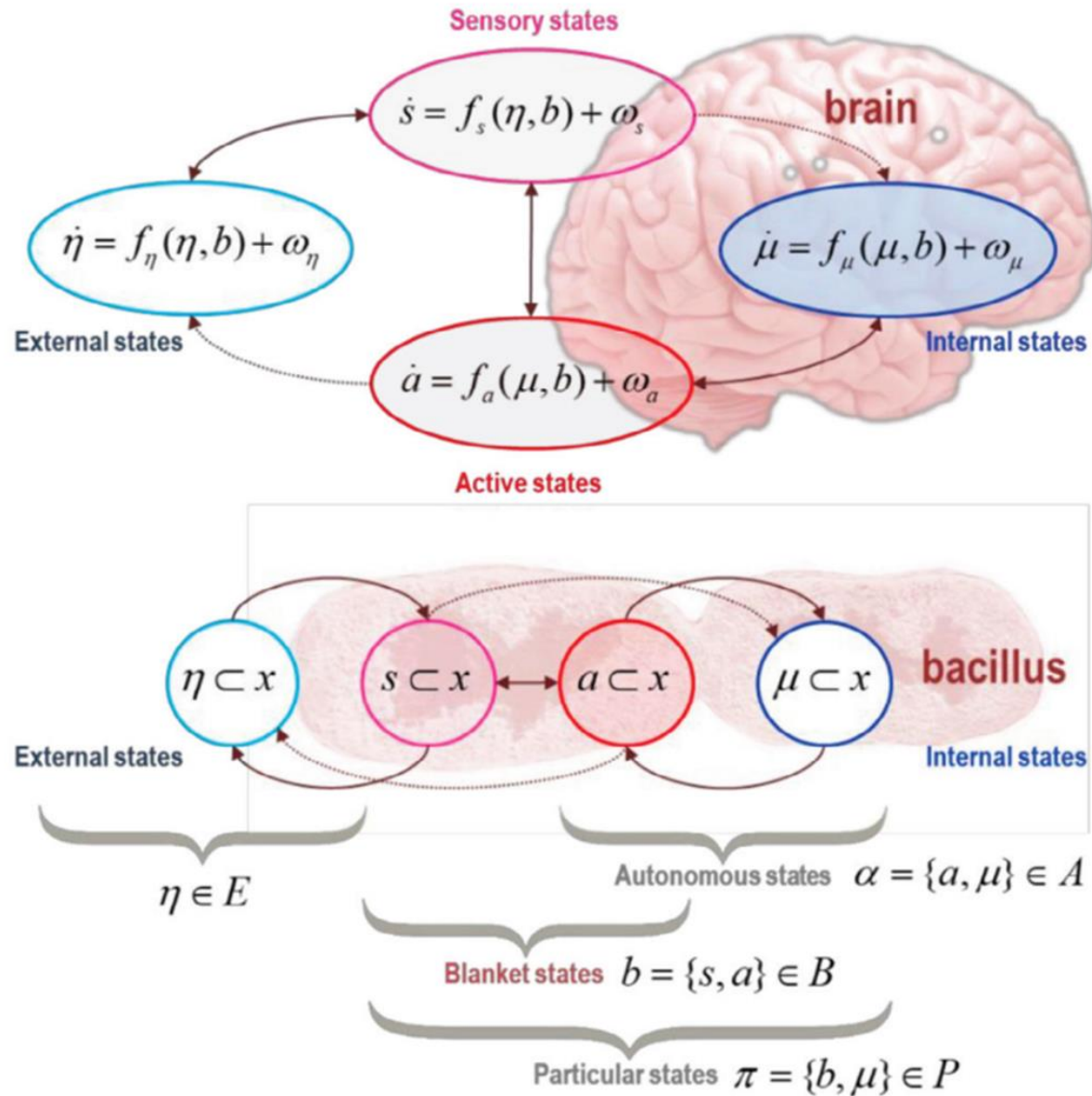


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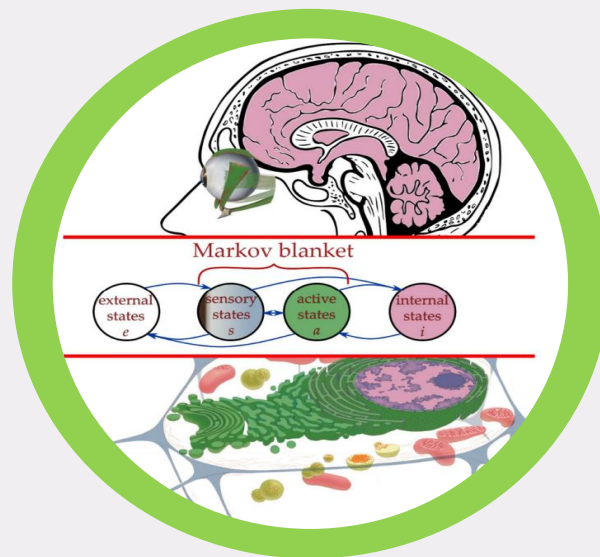


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Markov Blanket in the Brain

A MB is a statistical tool that allows explaining how an embodied system, situated in its environment, maintains itself by engaging in dynamics of exchanging with its environment



MB in the brain

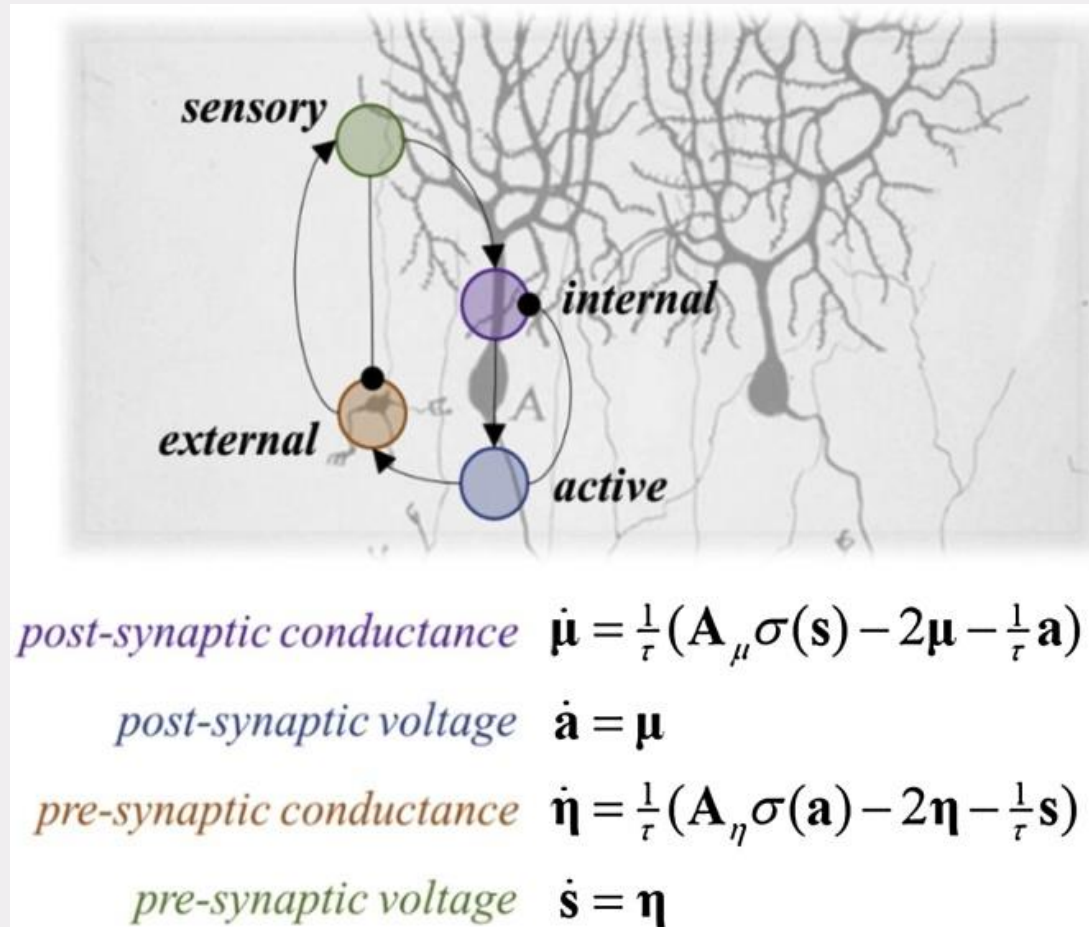
teams

timeline

history

about

Neurons and their Markov blankets



MB in the brain

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References

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